

health and performance of networks, servers and applications, through a single integrated software package. Written in the Python programming language, it also uses the Zope application server, and MySQL for data storage.

Installation

Differing from the typical cryptic installation procedures for many Linux-based networking tools, which may include manual dependency resolution, various command-line options, making the required changes in OS modules, and more, Zenoss installation is an utter delight—just a double-click! All you do is install CentOS 5.2 (or any Linux OS of your choice), then download (from <http://community.zenoss.org/community/download>) and double-click the `zenoss-stack-3.1.0-linux.bin` installer. The minimum hardware requirements are 2 GB of RAM and 20 GB of free disk space.

Configuration

Once installed, start configuring Zenoss using its GUI, which is accessible on `localhost:8080`. To enable it to be accessed from the network, you have to modify CentOS' `iptables` firewall settings to open port 8080, as follows:

1. Edit the `iptables` file with `vi /etc/sysconfig/iptables`, or your favourite editor.
2. Append the following rule:

```
-A RH-Firewall-1-INPUT -m state --state NEW -m tcp -p tcp
--dport 8080 -j ACCEPT
```

3. Save and close the file, then restart the `iptables` service with `service iptables restart`.
4. You can verify whether the port is open using the following command:

```
#netstat -tulpn | grep 8080
tcp      0      0 0.0.0.0:8080          0.0.0.0:*
LISTEN   2751/.python.bin
```

With the basic installation and configuration done, let's look at how to configure important Zenoss features, with screen-shots. The default Zenoss page has five important tabs, as seen in Figure 1. These are:

- **Dashboard:** This gives a consolidated picture of the monitored devices.
- **Events:** Details various events, sorted in terms of criticality.
- **Infrastructure:** Lists all monitored devices, lets you add/remove devices and set various parameters. The sub-menus also provide networks, various service classes and a network map for SNMP-enabled devices, among other functionality.
- **Reports:** Provides a detailed status of the devices, including the number of errors and the percentage of



Figure 1: Main menu

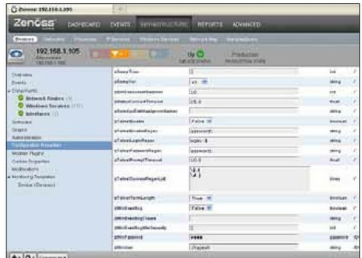


Figure 2: Infrastructure, device configuration

their resources that are available.

- **Advanced:** Various configuration settings, including the email address to send an alert to in case certain specified events occur, and the addition of Zenpacks (additional modules not available in the default Zenoss).

Now, let us consider some of the important menus in more detail.

Infrastructure

To add a host, device or service to the monitored list, specify the details here. The item can be classified as a router, host or service; whether it is a production or test unit, and more. For example, let's consider a scenario where we add an MS Windows XP-based desktop, on which some important services need to be monitored. (For host monitoring, a ping response may be sufficient, but for services monitoring, the system must be SNMP-enabled.) Configuring the desktop to enable SNMP is a two-step process:

Go to **Control Panel > Add/Remove Programs > Add/Remove Windows Component > Management and Monitoring Tools**. Tick the 'Simple Network Management Protocol' and 'WMI SNMP Provider' boxes. Click OK, and insert the Windows CD, if required.

Install agent software to monitor and manage the Windows machine—to maintain its Management Information Base (MIB), and respond to requests from network-monitoring systems (in this case, Zenoss). SNMP Informant is an excellent candidate; the standard version is available for free download, while the premium version requires payment.